

JAVA PROGRAMMING ASSIGNMENT

1. HEADER

Name: Ponnaganti Jaswanth Naidu

Roll No.: 192524322

Course Code & CO Reference: CSA0920, CO1

Assignment Set: Scenario-Based

Question Number: Q27

Assigned Question

Q27. Counting Word Occurrences with a Map

Write a Java program that accepts a sentence or paragraph from the user and counts how many times each distinct word appears in it, using a **HashMap** where the key is the word and the value is its running count. Display each word along with its final occurrence count.

Concepts: HashMap<String, Integer>, Word Frequency.

GitHub Repository Link:

<https://github.com/jaswanth-1503/Java/commit/d9e23936d4c5e6149e30ffea3b0c0d099a40481e>

Submission Date: 31 / 07 / 2026

2. PROBLEM UNDERSTANDING

The aim of this program is to count the number of occurrences of each word in a sentence or paragraph entered by the user. The program uses a **HashMap** to store unique words as keys and their occurrence counts as values. This assignment helps in understanding Java Collections, string manipulation, loops, and the HashMap data structure.

3. APPROACH / DESIGN NOTES

- Used the **HashMap<String, Integer>** class to store words and their frequencies.

- Converted the input into lowercase so that uppercase and lowercase words are treated equally.
 - Split the sentence into individual words using the `split("\\s+")` method.
 - Used the `containsKey()` method to check whether a word already exists in the `HashMap`.
 - Displayed the word frequencies using `Map.Entry`.
-

4. SOURCE CODE

```
import java.util.*;

public class WordCount {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a sentence:");

        String sentence = sc.nextLine();

        HashMap<String, Integer> map = new HashMap<>();

        String[] words = sentence.toLowerCase().split("\\s+");

        for (String word : words) {

            if (map.containsKey(word)) {

                map.put(word, map.get(word) + 1);

            } else {

                map.put(word, 1); }

        }

        System.out.println("\nWord Occurrences:");

        for (Map.Entry<String, Integer> entry : map.entrySet()) {

            System.out.println(entry.getKey() + " : " + entry.getValue());

        }

        sc.close();

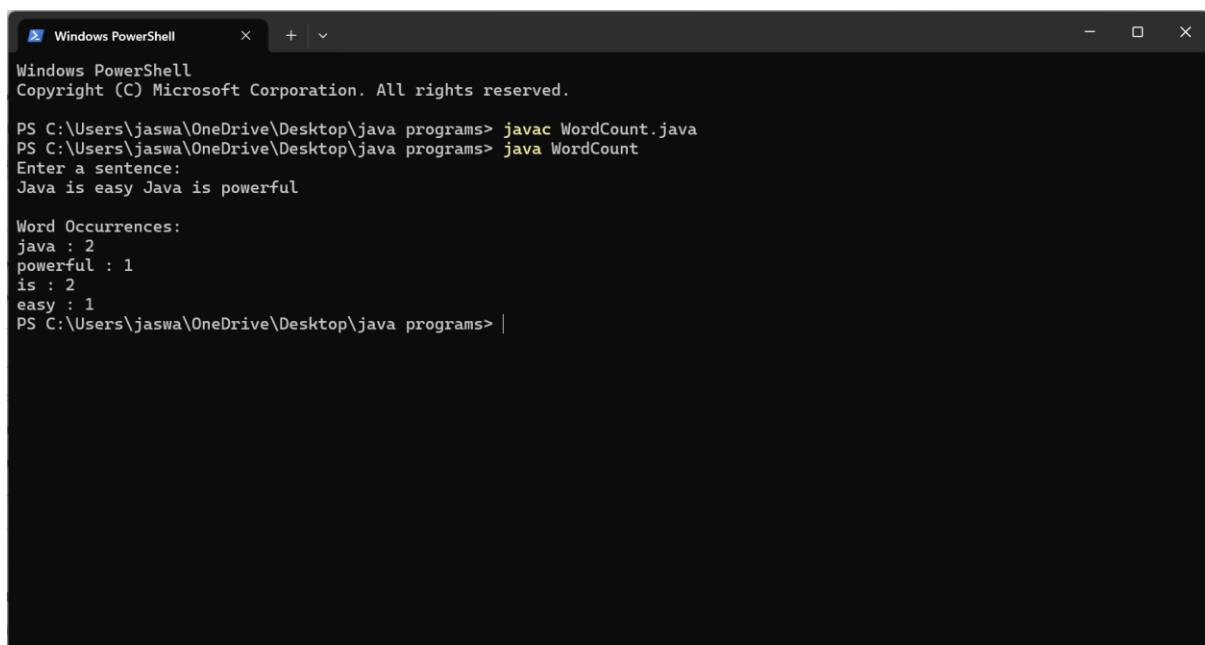
    }

}
```

5. TEST CASES

Test Case	Input	Expected Output	Actual Output
1 (Typical)	Java is easy Java is powerful	java : 2, is : 2, easy : 1, powerful : 1	Same as Expected

6. SCREENSHOTS OF EXECUTION



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\jaswa\OneDrive\Desktop\java programs> javac WordCount.java
PS C:\Users\jaswa\OneDrive\Desktop\java programs> java WordCount
Enter a sentence:
Java is easy Java is powerful

Word Occurrences:
java : 2
powerful : 1
is : 2
easy : 1
PS C:\Users\jaswa\OneDrive\Desktop\java programs> |
```

7. REFLECTION

This assignment helped me understand how to use a HashMap to count word frequencies efficiently. I also learned how to process strings, use loops, and apply Java collection classes to solve real-world problems.

8. DECLARATION

"I hereby declare that this assignment is my own work and has been completed by me. It has not been copied from any other source and is submitted for academic purposes."
